

SHARP

ELECTRONIC CALCULATOR

ELSI MATE

MODEL

EL-CC12D
EL-CC12G
EL-CC12GP
EL-CC14GP

OPERATION MANUAL

SHARP CORPORATION

20ESC(TINSEA306EHZZ)

ENGLISH

BEFORE USE

SPECIFICATIONS

• Do not press too hard against the LCD panel because it contains glass.

• Never dispose of the battery in a fire.

• Keep battery out of reach of children.

• Since this product is not waterproof, do not use it or store it where fluids, for example water, can splash onto it. Raindrops, water spray, juice, coffee, steam, perspiration, etc. will also cause malfunction.

• Please press **C/CE** if you see no indication.

• This product, including accessories, may change due to upgrading without prior notice.

SHARP will not be liable nor responsible for any incidental or consequential economic or property damage caused by misuse and/or malfunctions of this product and its peripherals, unless such liability is acknowledged by law.

Type: Electronic calculator

Operating capacity:
[EL-CC12D/EL-CC12G/CC12GP] 12 digits
[EL-CC14GP] 14 digits

Power supply: Built-in solar cell and Alkaline manganese battery (1.5V $\times$; (DC) LR44 or equivalent $\times$ 1)

Operating time: Approx. 2 years
(varies according to use and other factors)
(with 1 hour of daily use, using the alkaline manganese battery only)

Automatic Power-off: Approx. 7 min.

Operating temperature: 0°C - 40°C (32°F-104°F)

Dimensions:
[EL-CC12D] 132 mm(W) $\times$ 166 mm(D) $\times$ 30 mm(H)
[EL-CC12G/EL-CC12GP] 132 mm(W) $\times$ 166 mm(D) $\times$ 30 mm(H)
[EL-CC14GP] 158 mm(W) $\times$ 198 mm(D) $\times$ 32 mm(H)

Weight:
(battery included)
[EL-CC12D] Approx. 169 g
[EL-CC12G] Approx. 236 g
[EL-CC12GP] Approx. 240 g
[EL-CC14GP] Approx. 244 g

Accessories: Alkaline manganese battery (installed), Operation manual

OPERATIONS

1. Before starting calculations, press **CA** to clear (zero-clear).

2. The position of any switch that is not specifically designated, may be set to any position.

3. For expressing calculation examples, only the symbols that are required for explanation are mentioned.

4. Example procedures are listed in following manner unless otherwise specified.

(1) Example

(2) Key operations

(3) Display

F 4 3 2 0 A

DECIMAL POINT ASSIGNMENT SWITCH
"F" position: Set the decimal point to "floating" point.
"4,3,2,0" positions: The decimal figures can be set to the respective numbers.
"A" position: Assigns Adding mode (sets the decimal at 2). In the case of addition/subtraction, the decimal point will be placed at 2 automatically. If is registered, or in case of non-additive/non-subtractive calculation, this function will not be applicable.

t 5/4

ROUNDING SELECTOR
Specifies the handling of the digit immediately below the position set by the decimal selector.
(For example, if the decimal selector is set to 3, then the fourth decimal will be rounded.)

↑ 5/4

"↑" position: The answer will be rounded up.
"5/4" position: The answer will be rounded off.
"↓" position: The answer will be rounded down.

* While the decimal selector is set to the "F" (i.e., floating) position, or if the decimal place cannot be determined (preliminary calculation results, for instance), the answer will always be rounded down regardless of the setting above.

STATUS DISPLAY AND COUNTER

(1) $23 \times 2 \div 4 = 11.5$

(EL-CC12GP/EL-CC14GP: F 4 3 2 0 A)

(2)

(3)

C/CE C/CE

23

$\times$

2

001

$\times$

002

0.

23.

23.

2.

$\div$

4

$=$

002

003

004 G

46.

4.

11.5

CHECK & CORRECT

• The maximum number of calculation steps (operators and numerical values) that can be stored is 300. And when 300 steps are reached, correction/insertion/deletion cannot be performed.

• Symbols when replaying may be different from those when calculating
(ex. "M" symbol of memory calculations is displayed as recall memory or clear memory when replaying.).

• In some calculations, correction/insertion/deletion cannot be performed.

• Pressing **CA** will clear all memories, including stored calculation steps. And if the battery is consumed largely, all memories are also cleared.

• At the time of correction/insertion/deletion, if decimal point settings or rounding settings or tax rate of stored each calculation step are changed, all stored calculation steps will be recalculated with the changed settings.

• After starting correction/insertion (pressing **CORRECT** / **INSERT**), the entered value is canceled by pressing **C/CE** **C/CE** before pressing **CORRECT** / **INSERT** to set the entered value.

• If an error occurs ("E" symbol appears), all stored calculation steps will be cleared.

• "BUSY" may be displayed during recalculation.

AUTO REPLAY

• Auto Replay starts by pressing **AUTO REPLAY** and you can recall a calculation formula automatically.

Auto Replay stops by pressing **AUTO REPLAY** during Auto Replay and pressing **AUTO REPLAY** again starts Auto Replay to resume.

(1) $(400 - 80 + 500) \times 3 = 2460$

(EL-CC12GP/EL-CC14GP: F 4 3 2 0 A)

(2)

(3)

C/CE C/CE

400

$-$

80

$+$

500

$\times$

3

$=$

AUTO REPLAY

005 G

001 AUTO REPLAY

002 AUTO REPLAY

$=$

2'460.

400.

80.

PREV NEXT

PREVIOUS / NEXT

• Each time **NEXT** is pressed, calculation formulas are recalled in the order they are entered, while each time **PREV** is pressed, calculation is recalled in the order of its answer.

(1) $(400 - 80 + 500) \times 3 = 2460$

(EL-CC12GP/EL-CC14GP: F 4 3 2 0 A)

(2)

(3)

C/CE C/CE

400

$-$

80

$+$

500

$\times$

3

$=$

PREV

PREV

PREV

005 G

005 REPLAY

004 REPLAY

2'460.

2'460.

3.

CORRECT

(1) $(400 - 80 + 500) \times 3 \rightarrow (400 - 100 + 500) \times 3 = 2400$

(EL-CC12GP/EL-CC14GP: F 4 3 2 0 A)

(2)

(3)

C/CE C/CE

400

$-$

80

$+$

500

$\times$

3

$=$

NEXT

NEXT

CORRECT

005 G

001 REPLAY

002 REPLAY

002 REPLAY CORRECT

002 REPLAY CORRECT

2'460.

400.

80.

80.

100.

CORRECT

NEXT

NEXT

NEXT

(AUTO REPLAY)

$\downarrow$

002 REPLAY

003 REPLAY

004 REPLAY

005 REPLAY

100.

500.

3.

2'400.

INSERT

(1) $(400 - 80 + 500) \times 3 \rightarrow (400 - 80 + 500) \div 2 \times 3 = 1230$

(EL-CC12GP/EL-CC14GP: F 4 3 2 0 A)

(2)

(3)

C/CE C/CE

400

$-$

80

$+$

500

$\times$

3

$=$

PREV

PREV

INSERT

$\div$

005 G

005 REPLAY

004 REPLAY

004 REPLAY INSERT +

004 REPLAY INSERT $\div$

2'460.

2'460.

3.

0.

0.

2

INSERT

NEXT

NEXT

$\downarrow$

004 REPLAY INSERT $\div$

004 REPLAY $\div$

005 REPLAY $\times$

006 REPLAY

2.

2.

3.

1'230.

DEL/→ DELETE

(1) $(400 - 80 + 500) \times 3 \rightarrow (400 + 500) \times 3 = 2700$

(EL-CC12GP/EL-CC14GP: $\overset{F4320A}{\text{U}}$)

(2)	(3)
C-CE C-CE 400 - 80 + 500 × 3 = NEXT NEXT	005 G 2'460. 400. 80.

DEL/→	002 REPLAY +	500.
NEXT	003 REPLAY ×	3.
NEXT	004 REPLAY =	2'700.

• To check a calculation formula after performing the Correct, Insert and Delete function, press $\overset{PREV}{\blacktriangle}$ / $\overset{NEXT}{\blacktriangledown}$ repeatedly or press $\overset{AUTO}{\text{REPLAY}}$.

MARKUP CALC.

(1)	(2)	(3)
Cost = \$350, Markup % = 30% → Selling Price ?(500), Markup ?(150)	C-CE C-CE 350 ÷ 30 MU MU	MU 500. MU 150.
Selling Price = \$500, Markup % = 30% → Cost ?(350)	C-CE C-CE 500 × 30 +/- MU	MU 350.
Cost = \$350, Markup % on cost = 30% → Selling Price ?(455)	C-CE C-CE 350 × 30 MU	MU 455.
Selling Price = \$455, Markup % on cost = 30% → Cost ?(350), Markup ?(105)	C-CE C-CE 455 ÷ 30 +/- MU MU	MU 350. MU 105.

CALCULATING TAX

• To perform a tax calculation, first it is absolutely necessary to set and check the tax rate.
(The initial tax rate is 0.)
A tax rate can be set (or changed) with a number containing up to four digits. (Decimal point is not regarded as a digit.)
The set tax rate is retained until it is changed. However, the set tax rate may be changed or lost if the battery is consumed largely.

(1) • Confirming the tax rate. (0%)

(2)	(3)
C-CE C-CE RATE RECALL -TAX	TAX % 0.

(1) • Set a 5% tax rate.
Calculate the tax on \$800 and calculate the total Including tax.

(2)	(3)
C-CE C-CE RATE SET +TAX 5 RATE SET +TAX	TAX % = 5.
C-CE C-CE 800 +TAX	800. +TAX 840.
+TAX	TAX 40.

(1) • Perform two calculations using \$84 and \$52.5, both of which already include tax. (tax rate: 5%)
Calculate the tax on the total and the total without tax.

(2)	(3)
C-CE C-CE 84 + 52.5 = -TAX -TAX	136.5 -TAX 130 TAX 6.5

HOW TO PERFORM BASIC CALCULATIONS

(1)	(2)	(3)
	(EL-CC12GP/EL-CC14GP: $\overset{F4320A}{\text{U}}$)	
	GA	0.
$(-24+2) \div 4 = -5.5$	24 +/- 2 ÷ 4 =	-5.5
$13 \times (-4) + 2 = -26$	13 × 4 +/- ÷ 2 =	-26.
$34 + 57 = 91$	34 + 57 =	91.
$45 + 57 = 102$	45 =	102.
$38 - 26 = 12$	38 - 26 =	12.
$35 - 26 = 9$	35 =	9.
$68 \times 25 = 1700$	68 × 25 =	1'700.
$68 \times 40 = 2720$	40 =	2'720.
$35 \div 14 = 2.5$	35 ÷ 14 =	2.5
$98 \div 14 = 7$	98 =	7.
$200 \times 10\% = 20$	200 × 10 % =	20.
$(9 \div 36) \times 100 = 25$	9 ÷ 36 % =	25.
$200 + (200 \times 10\%) = 220$	200 + 10 % =	220.
$4^6 = (4^3)^2 = 4096$	4 × = = × =	4'096.
$1/8 = 0.125$	8 +/- =	0.125
$25 \times 5 = 125$ $-) 84 \div 3 = 28$ $+) 68 + 17 = 85$ 182	RCM RCM 25 × 5 (M+) 84 ÷ 3 (M-) 68 + 17 (M+) RCM RCM	M 125. M 28. M 85. M 182. 182.
$2 + 3 \rightarrow 2 + 4 = 6$	2 + 3 (C-CE) 4 =	6.
$5 \times 2 \rightarrow 5 \div 2 = 2.5$	5 × ÷ 2 =	2.5
$123456 \rightarrow 123478$	123456 DEL/→ DEL/→ 78	123'478.
$23.5 + 42.3 = 65.8$ $72.4 - 37.32 + 12 = 47.08$ $+) 27 - 19 - 21.61 = -13.61$ 99.27	(GT) (GT) C-CE 23.5 + 42.3 = 72.4 - 37.32 + 12 = 27 - 19 - 21.61 = (GT)	0. 65.8 47.08 -13.61 99.27
EL-CC12D / EL-CC12G / EL-CC12GP: $987654320988 \div 0.444$ $\times 555$ $= 1234.56790123 \times 10^{12}$	C-CE 987654320988 ÷ 0.444 × C-CE 555 = (1234.56790123 × 10¹² = 1234567901230000)	2.22444666889 1'234.56790123
EL-CC14GP: $98765432099998 \div 0.444$ $\times 555$ $= 1234.5679012499 \times 10^{14}$	C-CE 98765432099998 ÷ 0.444 × C-CE 555 = (1234.5679012499 × 10¹⁴ = 123456790124990000)	2.2244466689188 1'234.5679012499
EL-CC12GP / EL-CC14GP: $\overset{F4320A}{\text{U}}$	C-CE 145 + 4500 - 763 =	1.45 46.45 38.82

Note: $\square$ was not used in the entries.

BATTERY REPLACEMENT

